

Nextflow from zero to hero

Welcome to the course



Luigi



Youssef



Edoardo



Daniel



Bruno



Matteo

Expectations

What's your current experience using Nextflow?

How do you plan to use Nextflow in your work?

What are your expectations from this training course?



Course Organization



01 Theoretical concepts will be clearly explained through slides

03 Strong emphasis placed on hands-on practical implementation

02 Learning will be interleaved with focused practical exercises

04 Interactive sessions encourage active participation and learning

Course materials

Main repository



<https://github.com/HTGenomeAnalysisUnit/nextflow-zero2hero.git>

Folder on HPC



`/project/nextflow-zero2hero`



Example data

`/project/nextflow-zero2hero/data`



Code tutorials

`/project/nextflow-zero2hero/practicals`



Course Agenda: At a Glance

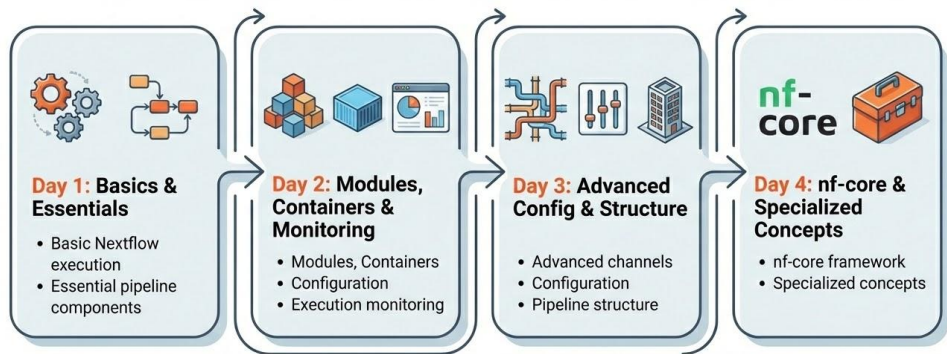
- **Day 1**
focuses on basic Nextflow execution and essential pipeline components
- **Day 2**
covers modules, containers, configuration, and execution monitoring
- **Day 3**
explores advanced channels, configuration, and pipeline structure
- **Day 4**
introduces the nf-core framework and its specialized concepts



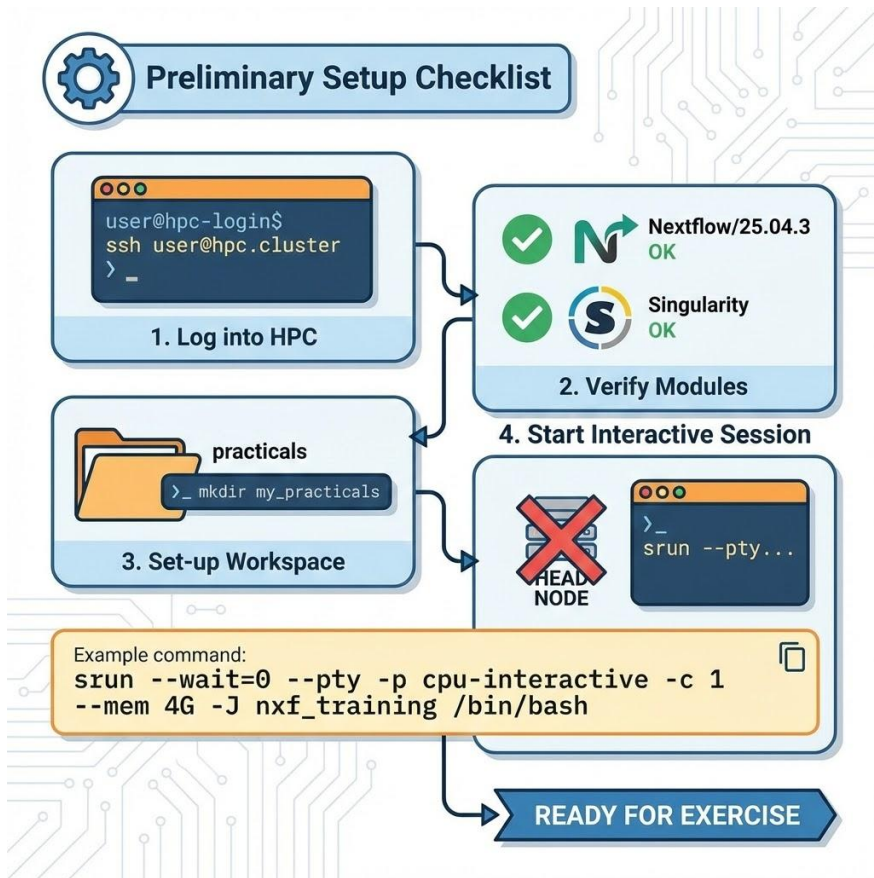
Daily: 9:00 - 17:00



Lunch Break: 13:00 (1 hr)



Setup your environment



Practicals welcome README:

<https://github.com/HTGenomeAnalysisUnit/nextflow-zero2hero/practicals/README.md>

Useful resources

nextflow

- Nextflow web page

<https://www.nextflow.io/>



- Nextflow main documentation

<https://www.nextflow.io/docs/stable/index.html>



- **Sequera AI** (Nextflow-specialized agent)

<https://sequera.io/ask-ai/chat-v2>

nf-core



- Nf-core pipelines

<https://nf-co.re/pipelines>



- Sequera Nextflow community

<https://community.sequera.io/tag/nextflow>